Introduction

Air Quality Index (AQI) is a numerical scale used to represent the quality of air and its impact on human health. It helps in understanding how polluted the air is and what associated health effects might be of concern. Higher AQI values indicate worse air quality and increased health risks. Monitoring AQI is essential for environmental protection and public health awareness.

Dataset Description:

The dataset used for this analysis was obtained from Kaggle. It contains approximately 1400 rows and 12 columns representing daily air quality measurements.

The dataset includes:

- Date, Month, Year
- Pollutants: PM2.5, PM10, NO2, SO2, CO, Ozone
- AQI (Air Quality Index)

The AQI values were already provided in the dataset and used directly for analysis.

Date	Month	Year	Holidays_Count	Days	PM2.5	PM10	NO2	SO2	CO	Ozone	AQI	Formatted Dates
1	1	2021	0	5	180.57	458.9	42.35	10.03	1.57	19.04	336	01 January 2021
2	1	2021	0	6	146.71	371.52	27.41	10.05	1.14	13.37	351	02 January 2021
3	1	2021	1	7	76.94	190.39	13.17	10.32	0.96	14.67	241	03 January 2021
4	1	2021	0	1	58.12	111.38	67.6	12.79	1.01	12.77	90	04 January 2021
5	1	2021	0	2	44.66	86.96	55.96	9.97	1.41	15.82	99	05 January 2021
6	1	2021	0	3	74.92	158.51	24.95	10.87	1.14	18.63	125	06 January 2021
7	1	2021	0	4	68.94	158.37	21.69	9.63	1.22	17.04	164	07 January 2021
8	1	2021	0	5	67.81	154.24	15.97	10.32	1.3	17.07	125	08 January 2021
9	1	2021	0	6	82.81	197.12	18.89	10.39	1.18	21.45	182	09 January 2021
10	1	2021	1	7	72.44	152.12	14.32	9.53	1	16.58	148	10 January 2021

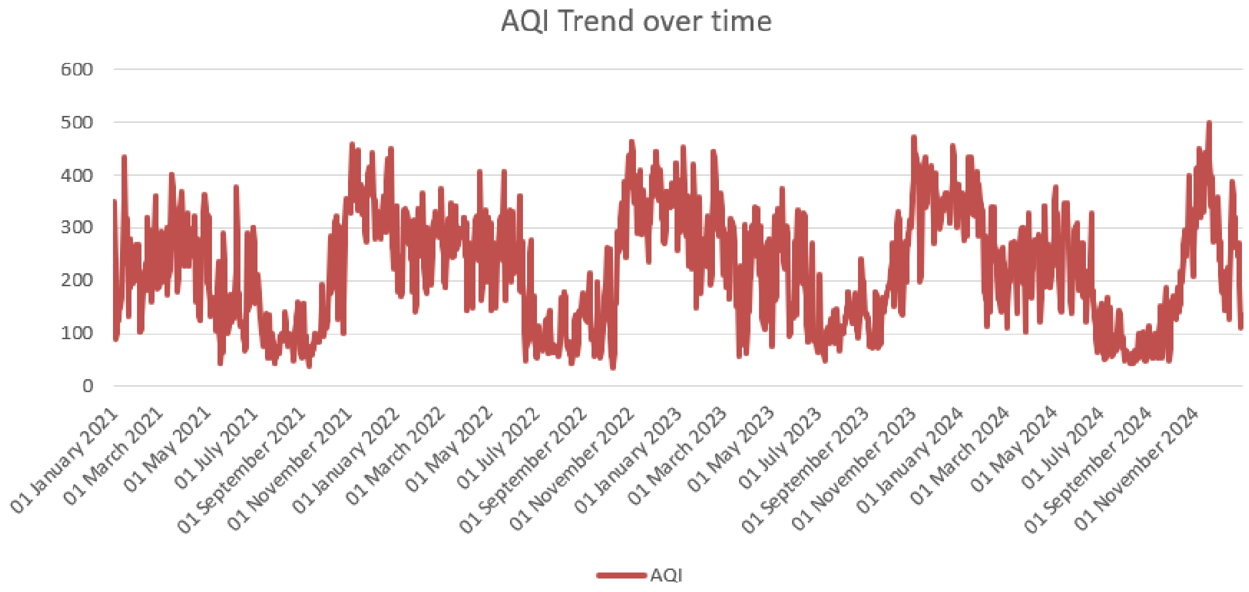
Data Cleaning and processing

The dataset was examined for missing values and duplicate entries. No major inconsistencies were found. Basic preprocessing steps such as ensuring correct data types and formatting (especially for date-related columns) were performed. The data was considered suitable for analysis.

Data Analytics and Visualization

1. AQI Trend over time

A line graph was created to visualize how AQI changes over time



Observations:

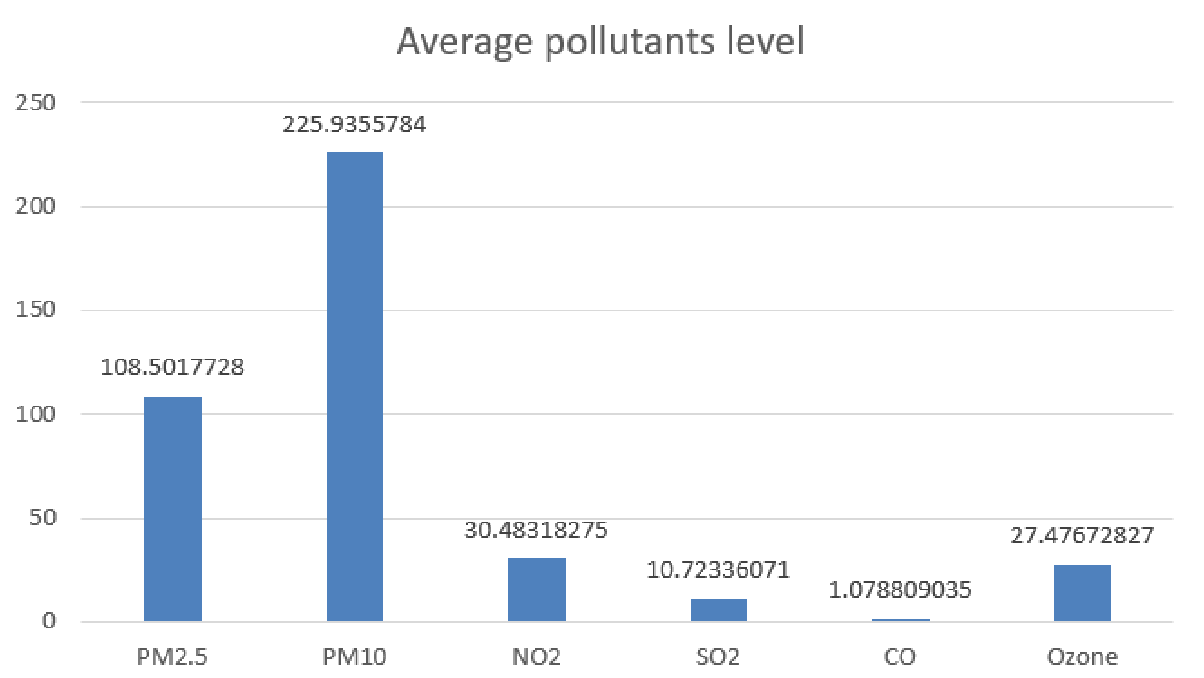
- AQI values fluctuate significantly over time
- Several peaks indicate high pollution levels
- Some AQI values exceed 300, indicating very poor to severe air quality

2. Average pollutants level

Average pollutants were calculated based on the given data

Pollutant	Value
PM2.5	108.5017728
PM10	225.9355784
NO2	30.48318275
SO2	10.72336071
CO	1.078809035
Ozone	27.47672827

A bar chart was created to compare the average concentration of pollutants



Observations:

- PM10 has the highest average value followed by PM2.5
- These pollutants are the major contributors to air pollution
- Other pollutants such as SO2 and CO have comparatively lower levels

AQI Assessment based on Standards

The AQI values were interpreted using standard air quality categories

AQI Range	Category
0-50	Good
51-100	Satisfactory
101-200	Moderate
201-300	Poor
301-500	Very Poor/ Severe

Based on the analysis, many AQI values fall under the Poor to Very Poor categories, indicating unhealthy air quality conditions.

Findings

- Air quality shows significant variation over time
- Several days exhibit very high AQI values
- PM2.5 and PM10 are the dominant pollutants

- High AQI levels indicate potential health risks

Conclusion

The analysis highlights that air pollution levels are frequently high and can pose serious health concerns. Monitoring AQI and controlling major pollutants such as PM2.5 and PM10 is essential for improving air quality. This study demonstrates the importance of continuous air quality monitoring and effective environmental management strategies

References

1. Kaggle – Delhi Air Quality dataset. Available at [Delhi Air Quality Dataset]
2. Central Pollution Control Board (CPCB), India – AQI Standards
3. World Health Organization (WHO) – Air Quality Guidelines